FlatSat platforms for small satellites: systematic mapping

J. C. E. Barcellos

This document lists the works analyzed in the systematic mapping published in J. C. E. Barcellos *et al.*, "FlatSat Platforms for Small Satellites: A Systematic Mapping and Classification," *IEEE Journal on Miniaturization for Air and Space Systems*, vol. 4, no. 2, pp. 186–198, 2023, doi: 10.1109/JMASS.2023.3249044. It is generated from `data/flatsat_platforms.tsv` and `data/references.bib`, which may be expanded and corrected over time.

Summary

Type	Works
Raw	41
Bridge	6
Dock	18
Modular	3

Test	Works
Functional	61
Mission	14
Thermal	2
EMC	1

Platform characteristic	Works
Supporting structure (metallic or not)	34
Subsystems assembled on a backplane	27
Passive / active backplane	22 / 5
Sensors (e.g., power consumption, temperature)	1
Switches and/or jumpers	10
Connectors for external power supplies	12
Connectors, LEDs, and/or test points for monitoring	27
Other standardized connectors (e.g., USB, antenna)	6
CPLD	1

Total: 65 works. Works classified into more than one type (e.g., Raw/Dock) are counted in each.

Analyzed works

#[a]	Ref.	Mission / project	Year	Type	Assembly	Struct.	Backplane	Backplane features[b]						Tests			
								Sens.	Sw.	Pwr.	Mon.	Conn.	CPLD	Func.	Miss.	Therm.	EMC
1	[1]	EIRSAT-1	2021	Dock	Backplane	✓	Passive	✗	✗	✗	✓	✗	✗	✓	✓	✗	✗
2	[2]	SMDC TechSat	2012	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
4	[3]	3CAT-4	2021	Dock	Backplane	✗	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
5	[4]	Aalto-1	2021	Dock	Backplane	✓	Passive	✗	✓	✓	✓	✗	✗	✓	✗	✗	✗
6	[5]	CubeSat with a Raspberry Pi Zero	2021	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
7	[6]	ISTSat-1	2019	Modular	Backplane	✗	Passive	✗	✓	✓	✓	✗	✗	✓	✗	✗	✓
9	[7]	SMDC TechSat	2014	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
10	[8]	BA CubeSat	2021	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
11	[9]	OPS-SAT	2017	Raw/Dock	Both	✓	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗

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#[a]	Ref.	Mission / project	Year	Type	Assembly	Struct.	Backplane	Backplane features[b]						Tests			
								Sens.	Sw.	Pwr.	Mon.	Conn.	CPLD	Func.	Miss.	Therm.	EMC
12	[10]	QUBE	2018	Dock	Backplane	✗	Passive	✗	✓	✓	✓	✓	✗	✓	✗	✗	✗
13	[11]	Galassia	2016	Dock	Backplane	✗	Passive	✗	✗	✓	✓	✓	✗	✓	✓	✗	✗
14	[12]	DESCENT	2019	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
15	[13]	NTS CubeSat	2010	Raw	Wire	✗	-	-	-	-	-	-	-	✗	✓	✗	✗
16	[14]	NTS CubeSat	2009	Raw	Wire	✗	-	-	-	-	-	-	-	✗	✓	✗	✗
17	[15]	RACE, Bevo-2, and ARMADILLO	2014	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
18	[16]	FASTRAC	2010	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
19	[17]	VST104	2021	Dock	Backplane	✗	Active	✓	✓	✓	✓	✓	✗	✓	✗	✗	✗
20	[18]	MIST	2018	Bridge	Backplane	✗	Passive	✗	✓	✓	✓	✗	✗	✓	✗	✗	✗
22	[19]	EIRSAT-1	2021	Dock	Backplane	✓	Passive	✗	✗	✗	✓	✗	✗	✓	✓	✗	✗
23	[20]	FASTRAC	2008	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
24	[21]	EDSN	2014	Raw/Bridge	Both	✓	Passive	✗	✓	✗	✓	✗	✗	✓	✗	✗	✗
25	[22]	BIRDS-3	2019	Dock	Backplane	✗	Active	✗	✗	✗	✓	✗	✓	✓	✗	✗	✗
26	[23]	Galassia	2017	Dock	Backplane	✗	Passive	✗	✗	✗	✓	✗	✗	✓	✓	✗	✗
27	[24]	UWE-3	2015	Dock	Backplane	✗	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
29	[25]	GT-1	2021	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
31	[26]	Dellingr	2018	Bridge	Backplane	✓	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
33	[27]	CanX-4 and CanX-5	2011	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
34	[28]	Prox-1	2016	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
35	[29]	CanX-7	2014	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
36	[30]	MakerSat-1	2020	Raw	Wire	✓	-	-	-	-	-	-	-	✗	✗	✗	✗
38	[31]	Foresail-1	2020	Modular	Backplane	✗	Passive	✗	✗	✓	✓	✗	✗	✓	✗	✗	✗
39	[32]	MOVE-II	2018	Bridge	Backplane	✗	Passive	✗	✓	✗	✓	✗	✗	✓	✗	✗	✗
40	[33]	MOVE-II	2017	Bridge	Backplane	✗	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
41	[34]	CSSARSC	2019	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
42	[35]	MakerSat-1	2018	Raw	Wire	✓	-	-	-	-	-	-	-	✗	✗	✗	✗
43	[36]	Delfi-PQ-1	2018	Dock	Backplane	✗	Active	✗	✓	✓	✓	✓	✗	✓	✗	✗	✗
44	[37]	Prox-1	2016	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
45	[38]	CSSARSC	2020	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
46	[39]	POINTR	2019	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗

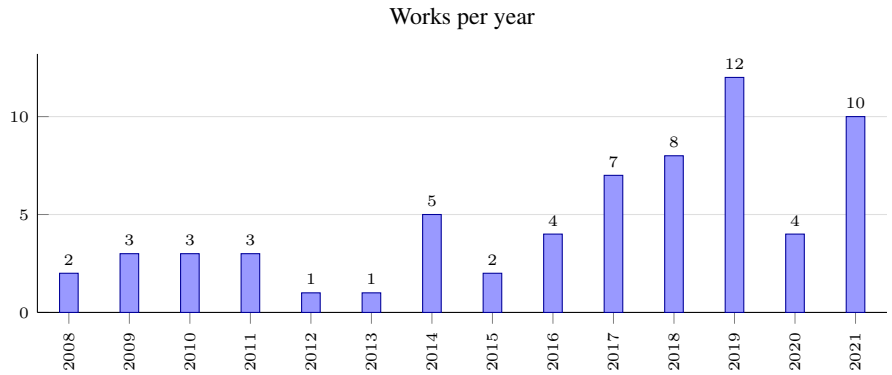
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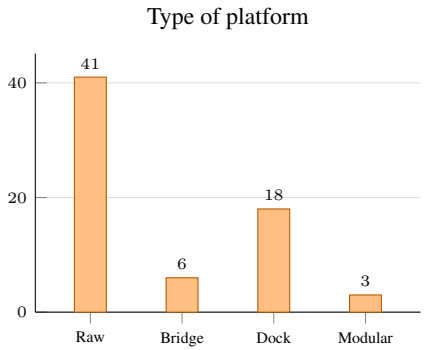
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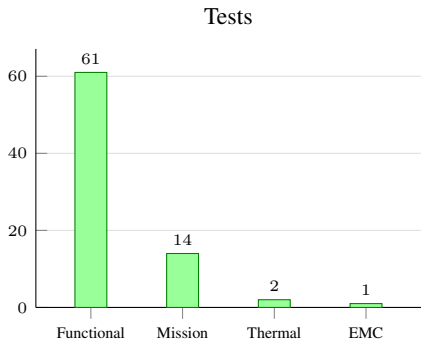
#[a]	Ref.	Mission / project	Year	Type	Assembly	Struct.	Backplane	Backplane features[b]						Tests			
								Sens.	Sw.	Pwr.	Mon.	Conn.	CPLD	Func.	Miss.	Therm.	EMC
47	[40]	NORSAT-2	2017	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
48	[41]	FASTRAC	2011	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
49	[42]	SSIVP	2017	Dock	Backplane	✗	Active	✗	✗	✓	✓	✓	✗	✓	✗	✗	✗
50	[43]	CanX-4 and CanX-5	2011	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
52	[44]	INTA-NanoSat-1B	2014	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
54	[45]	MOCI	2021	Dock	Backplane	✗	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
55	[46]	Nano-ADEPT	2019	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
56	[47]	NTS CubeSat	2008	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
59	[48]	CanX-2	2009	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
60	[49]	ESA-Prop	2019	Dock	Backplane	✗	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
61	[50]	SOAREX-8	2015	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✓	✗
63	[51]	CASPR	2021	Dock	Backplane	✗	Active	✗	✗	✓	✓	✓	✗	✓	✗	✗	✗
64	[52]	S.A.M.	2017	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
66	[53]	BRITE	2018	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
67	[54]	MIST	2019	Bridge	Backplane	✗	Passive	✗	✓	✓	✓	✗	✗	✓	✗	✗	✗
69	[55]	OPS-SAT	2019	Raw/Dock	Both	✓	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
70	[56]	UWE-3	2016	Dock	Backplane	✗	Passive	✗	✗	✗	✓	✗	✗	✓	✗	✓	✗
72	[57]	GHGSat-D	2017	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
73	[58]	SFL missions	2019	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
74	[59]	CHEOPS	2019	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
75	[60]	Flying Laptop	2020	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✗	✗	✗
76	[61]	TacSat-2	2010	Raw	Wire	✗	-	-	-	-	-	-	-	✓	✗	✗	✗
79	[62]	MarCO	2019	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
82	[63]	RACE, Bevo-2, and ARMADILLO	2013	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
83	[64]	SpaceDev	2009	Raw	Wire	✓	-	-	-	-	-	-	-	✓	✓	✗	✗
87	[65]	ISTSat-1	2018	Modular	Backplane	✗	Passive	✗	✓	✓	✓	✗	✗	✓	✗	✗	✗

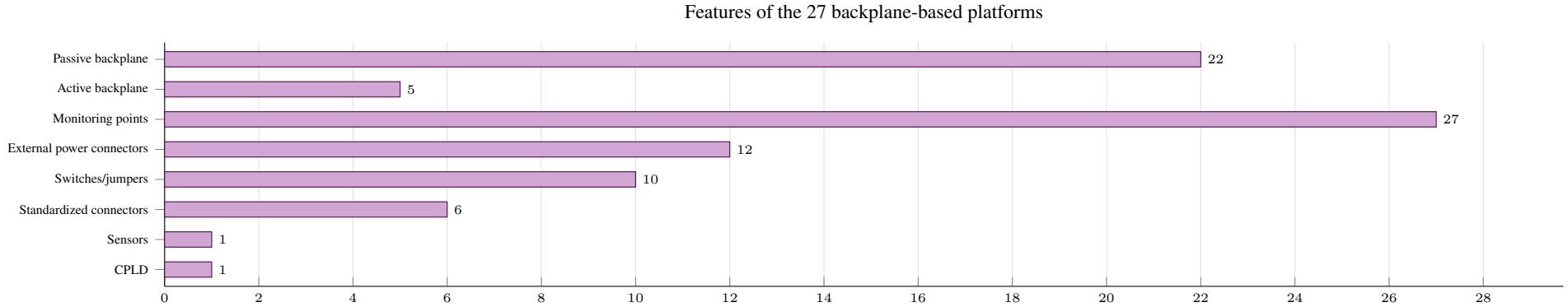
[a] Work number in the original screening.
[b] Backplane-based platforms only. Sens.: sensors; Sw.: switches and/or jumpers; Pwr.: connectors for external power supplies; Mon.: connectors, LEDs, and/or test points for monitoring; Conn.: other standardized connectors; CPLD: complex programmable logic device.
Note: -: not available or not applicable.

Charts









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